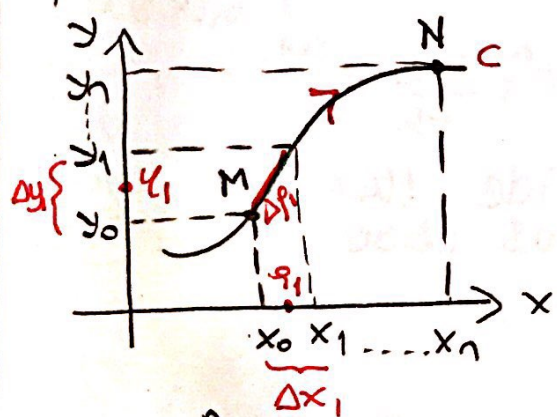


LINE INTEGRAL

Let consider the curve C given by parametric equations $x = x(t)$, $y = y(t)$ and $\widehat{MN}$ arc on this curve. Let the points M and N correspond to the parameters t_0 and t_n , respectively.

Suppose that the derivatives of the functions $x(t)$, $y(t)$ exist and continuous on $[t_0, t_n]$. Consider two functions $P(x, y)$ and $Q(x, y)$ such that they are continuous and have cont. derivatives at each point of $\widehat{MN}$ arc. Let cut $\widehat{MN}$ into parts, arbitrarily, and these points correspond to $t_0, t_1, \dots, t_n$. Choose arbitrary point in each part, $\mu_i(\xi_i, \eta_i)$



Arc $\Delta p_i \rightarrow \mu_i(\xi_i, \eta_i)$

$$\sum_{i=1}^n [P(\xi_i, \eta_i) \Delta x_i + Q(\xi_i, \eta_i) \Delta y_i]$$

let $n \rightarrow \infty$ to have
 $\Delta x_i \rightarrow 0$, $\Delta y_i \rightarrow 0$
 and $\Delta p_i \rightarrow 0$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n [P(\xi_i, \eta_i) \Delta x_i + Q(\xi_i, \eta_i) \Delta y_i]$$

if this limit exists then it's called the line integral along $\widehat{MN}$ arc. It is also defined as;

$$\int_{(M)}^{(N)} P(x, y) dx + Q(x, y) dy \quad \text{or} \quad \int_{\widehat{MN}} P(x, y) dx + Q(x, y) dy$$

Note: Line integral is defined as the work of the power that affects to a point moving along the curve C .

Similar definitions is also valid for the curves C in space.

$$\int_C P(x, y, z) dx + Q(x, y, z) dy + R(x, y, z) dz$$

Properties :

1) $\int_{(M)}^{(N)} P(x, y) dx + Q(x, y) dy = - \int_{(N)}^{(M)} P(x, y) dx + Q(x, y) dy$

2) If we have $\widehat{MN} = \widehat{MK} + \widehat{KN}$ then

$$\int_{(M)}^{(N)} P(x, y) dx + Q(x, y) dy = \int_{(M)}^{(K)} P(x, y) dx + Q(x, y) dy + \int_{(K)}^{(N)} P(x, y) dx + Q(x, y) dy$$

3) If the points M and N coincide then the curve C becomes closed curve. In this case, line integral is shown as :

$$\oint P(x, y) dx + Q(x, y) dy.$$

The positive direction is determined such that inside of the region will be on the left-hand side.

Vector Notation for Line Integrals

Line integral is defined as the work of $\vec{F}$ on a curve C . We can consider $\vec{F}$ as a vector valued function having the components $P(x, y)$, $Q(x, y)$.

[6]

Line integral of the vector function $\vec{F}$ along the curve C ;

$$\int_{(C)} P(x,y)dx + Q(x,y)dy = \int_{(C)} \vec{F} \cdot d\vec{r}$$

where $\vec{F} = P(x,y)\vec{i} + Q(x,y)\vec{j}$ and $d\vec{r} = dx\vec{i} + dy\vec{j}$

If the arc length s is considered;

$$\int \vec{F} \cdot d\vec{r} = \int \left(\vec{F} \cdot \frac{d\vec{r}}{ds} \right) ds = \int \vec{F} \cdot \vec{T} ds$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

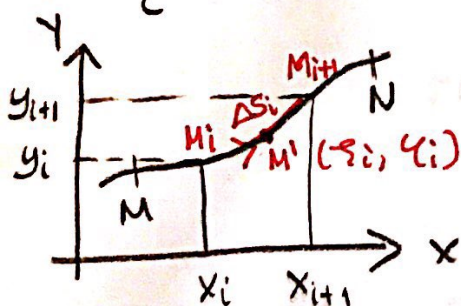
$$* \frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}$$

$$* \frac{d\vec{r}}{ds} = \vec{T} : \text{unit tangent vector}$$

$$* \frac{\frac{d\vec{r}}{dt}}{\left| \frac{d\vec{r}}{dt} \right|} = \frac{d\vec{r}}{ds} = \vec{T}$$

Calculation of Line Integral

$$I = \int_C P(x,y)dx + Q(x,y)dy \quad \text{and} \quad C : \begin{cases} x = g(t) \\ y = h(t) \end{cases}$$



$$\lim_{n \rightarrow \infty} \sum_{i=1}^n P(\xi_i, \eta_i) \Delta x_i + Q(\xi_i, \eta_i) \Delta y_i$$

$$\left. \begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=1}^n P(\xi_i, \eta_i) \Delta x_i &= \int_{\widehat{MN}} P(x, y) dx \\ \lim_{n \rightarrow \infty} \sum_{i=1}^n Q(\xi_i, \eta_i) \Delta y_i &= \int_{\widehat{MN}} Q(x, y) dy \end{aligned} \right\} \text{limits exist.}$$

These limits are ^{$\widehat{MN}$} not depending on the type of partition and choice of points.

$$x = g(t), y = h(t)$$

$$\Delta x_i = x_{i+1} - x_i = g(t_{i+1}) - g(t_i)$$

Use mean value theorem for $x = g(t)$.

$$\Delta x_i = g'(t_i')(t_{i+1} - t_i) = g'(t_i') \Delta t_i \quad t_i < t_i' < t_{i+1}$$

Choose $\xi_i = g(t_i')$, $\eta_i = h(t_i')$. In this case

$$\int_{\widehat{MN}} P(x, y) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n P[g(t_i'), h(t_i')] \cdot g'(t_i') \Delta t_i$$

As $n \rightarrow \infty \Rightarrow \Delta x_i \rightarrow 0 \Rightarrow \Delta t_i \rightarrow 0$ since they are cont. functions.
 $\Delta y_i \rightarrow 0$

$$\int_{\widehat{MN}} P(x, y) dx = \int_{t_0}^{t_n} P[g(t), h(t)] \cdot g'(t) dt$$

and similarly;

$$\int_{\widehat{MN}} Q(x, y) dy = \int_{t_0}^{t_n} Q[g(t), h(t)] \cdot h'(t) dt$$

$$\int_{MN} P(x, y) dx + Q(x, y) dy = \int_{t_0}^{t_1} \{ P[g(t), h(t)] g'(t) + Q[g(t), h(t)] h'(t) \} dt$$

A similar calculation is valid for line integral in space.

⊗ If curve C is defined with $y = f(x)$;

$$dy = f'(x) dx$$

$$\int_C P(x, y) dx + Q(x, y) dy = \int_{a_1}^{a_2} P(x, f(x)) dx + Q(x, f(x)) f'(x) dx$$

(a_1 and a_2 are on x -axis)

⊗ If curve C is defined with $x = g(y)$;

$$\int_C P(x, y) dx + Q(x, y) dy = \int_{b_1}^{b_2} P(g(y), y) g'(y) dy + Q(g(y), y) dy$$

(b_1 and b_2 are on y -axis)

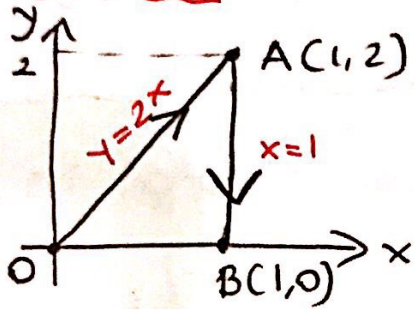
⊗ If parametric equations $\begin{matrix} x = g(t) \\ y = h(t) \end{matrix}$ are given;

$$x = g(t) \Rightarrow dx = g'(t) dt$$

$$y = h(t) \Rightarrow dy = h'(t) dt$$

$$\int_C P(x, y) dx + Q(x, y) dy = \int_{t_0(a_0, b_0)}^{t_1(a_1, b_1)} P(g(t), h(t)) g'(t) dt + Q(g(t), h(t)) h'(t) dt$$

Example: Evaluate $I = \int_C xy dx - x^2 y^2 dy$



$$C: \overline{OA} + \overline{AB}$$

$$I_1 + I_2$$

* $\overline{OA}$: $y=2x \Rightarrow dy=2dx$
 $x=x \Rightarrow dx=dx$

$$I_1 = \int_{\overline{OA}} x \cdot (2x) dx - x^2 \cdot 4x^2 \cdot 2dx \Rightarrow$$

$$I_1 = \int_0^1 (2x^2 - 8x^4) dx = \frac{2}{3}x^3 - \frac{8}{5}x^5 \Big|_0^1 = \frac{2}{3} - \frac{8}{5}$$

or: $y=y \Rightarrow dy=dy$
 $x=\frac{y}{2} \Rightarrow dx=\frac{dy}{2}$

$$I_1 = \int_0^2 \frac{y}{2} \cdot y \cdot \frac{dy}{2} - \frac{y^2}{4} \cdot y^2 dy = \int_0^2 \left(\frac{y^2}{4} - \frac{y^4}{4} \right) dy$$

$$I_1 = \frac{y^3}{12} - \frac{y^5}{20} \Big|_0^2 = \frac{8}{12} - \frac{32}{20} = \frac{2}{3} - \frac{8}{5}$$

* $\overline{AB}$: $x=1 \Rightarrow dx=0$
 $y=y \Rightarrow dy=dy$

$$I_2 = \int_{\overline{AB}} 1 \cdot y \cdot 0 - 1 \cdot y^2 dy = \int_2^0 -y^2 dy = \frac{y^3}{3} \Big|_2^0 = \frac{8}{3}$$

$$I = I_1 + I_2 = \frac{2}{3} - \frac{8}{5} + \frac{8}{3} = \frac{10}{3} - \frac{8}{5} = \frac{26}{15}$$

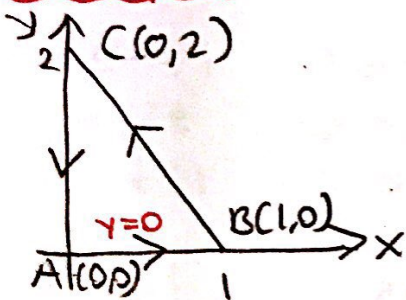
3rd way for $\overline{OA}$: $x=t \Rightarrow dx=dt$
 $y=2t \Rightarrow dy=2dt$

(x,y) $\rightarrow$ t
 $(0,0) \rightarrow 0$
 $(1,2) \rightarrow 1$

$$I_1 = \int_0^1 t \cdot 2t \cdot dt - t^2 \cdot 4t^2 \cdot 2dt$$

$$I_1 = \int_0^1 2t^2 dt - 8t^4 dt = \left[\frac{2}{3} t^3 - \frac{8}{5} t^5 \right]_0^1 = \frac{2}{3} - \frac{8}{5}$$

Example: Evaluate $I = \oint_C (x^3 - 3y)dx + (x + y^2)dy$
 $C : \overline{AB} + \overline{BC} + \overline{CA}$



* $\overline{AB}$: $y=0 \Rightarrow dy=0$
 $x=x \Rightarrow dx=dx$

$$I_1 = \int_{\overline{AB}} x^3 dx + 0 = \int_0^1 x^3 dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{4}$$

* $\overline{BC}$: $y=2-2x \Rightarrow dy=-2dx$
 $x=x \Rightarrow dx=dx$

$$I_2 = \int_{\overline{BC}} x^3 - 3(2-2x)dx + (x + (2-2x)^2)(-2dx)$$

$$I_2 = \int_1^0 [x^3 - 6 + 6x - 2x - 2(4 - 8x + 4x^2)] dx$$

$$I_2 = \int_1^0 (x^3 - 6 + 4x - 8 + 16x - 8x^2) dx$$

$$I_2 = \int_1^0 (x^3 - 8x^2 + 20x - 14) dx = \left[\frac{x^4}{4} - \frac{8}{3} x^3 + 10x^2 - 14x \right]_1^0$$

$$I_2 = 0 - \frac{1}{4} + \frac{8}{3} - 10 + 14 = -\frac{1}{4} + \frac{8}{3} + 4$$

* $\overline{CA}$: $x=0 \Rightarrow dx=0$
 $y=y \Rightarrow dy=dy$

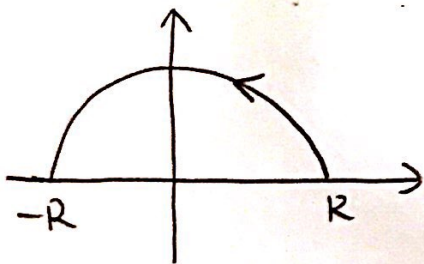
$$I_3 = \int_{\overline{CA}} -3y \cdot 0 + (0 + y^2) dy = \int_2^0 y^2 dy = \frac{y^3}{3} \Big|_2^0 = 0 - \frac{8}{3}$$

$$I = I_1 + I_2 + I_3$$

$$I = \frac{1}{4} + \frac{8}{3} - \frac{1}{4} + 4 - \frac{8}{3} = 4 //$$

Example: Evaluate

$$I = \int_C x dy - y dx$$



C : $\begin{cases} x = R \cos t \\ y = R \sin t \end{cases}$ parametric equations.

$$\begin{cases} x = R \cos t \Rightarrow dx = -R \sin t dt \\ y = R \sin t \Rightarrow dy = R \cos t dt \end{cases} \quad t: 0 \rightarrow \pi$$

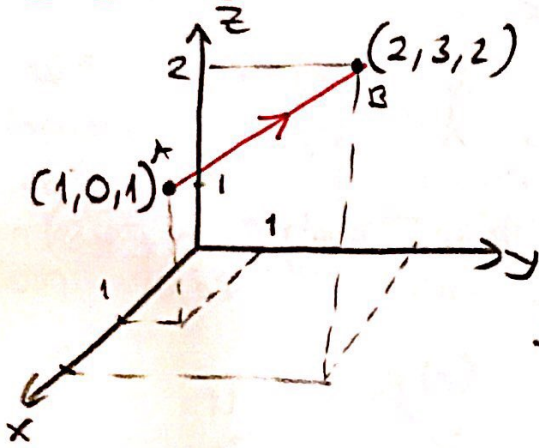
$$I = \int_0^\pi R \cos t \cdot R \cos t dt + R \sin t \cdot R \sin t dt$$

$$I = \int_0^\pi (R^2 \cos^2 t + R^2 \sin^2 t) dt = \int_0^\pi R^2 dt = R^2 \cdot t \Big|_0^\pi$$

$$\underline{I = R^2 \pi}$$

Example: Evaluate

$$I = \int_{(1,0,1)}^{(2,3,2)} x^2 dx - xz dy + y^2 dz$$



$$\vec{AB} = (1, 3, 1) \text{ direction of line } \ell$$

$$\ell: \begin{cases} x = 1+t \\ y = 0+3t \\ z = 1+t \end{cases} \text{ parametric equations of line } \ell.$$

$$\begin{aligned} x = t+1 &\Rightarrow dx = dt \\ y = 3t &\Rightarrow dy = 3dt \\ z = t+1 &\Rightarrow dz = dt \end{aligned}$$

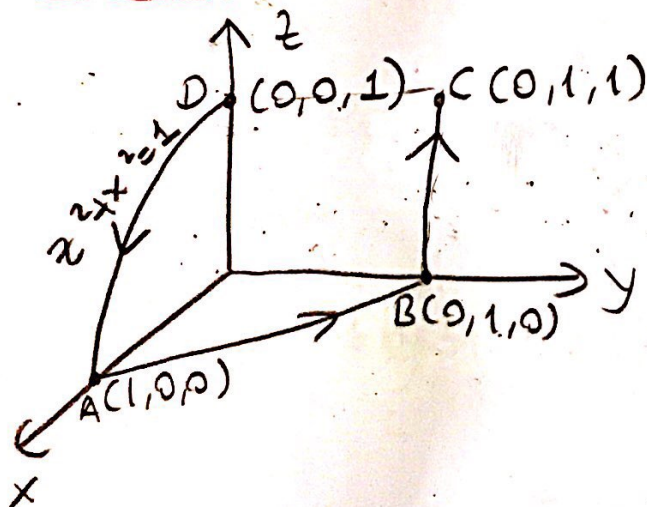
$$\begin{aligned} A(1,0,1) &\rightarrow t=0 \\ B(2,3,2) &\rightarrow t=1 \end{aligned}$$

$$I = \int_0^1 (t+1)^2 dt - (t+1)(t+1)3dt + 9t^2 dt$$

$$I = \int_0^1 (-2(t+1)^2 + 9t^2) dt = \int_0^1 (-2t^2 - 4t - 2 + 9t^2) dt$$

$$I = \int_0^1 (7t^2 - 4t - 2) dt = \left[\frac{7}{3}t^3 - 2t^2 - 2t \right]_0^1 = \frac{7}{3} - 4 = -\frac{5}{3} //$$

Example: $\vec{F} = xz\vec{i} + x\vec{j} - yz\vec{k}$ $\int_C \vec{F} d\vec{r} = ?$



$$C: \vec{DA} + \vec{AB} + \vec{BC}$$

$$* \widehat{DA} : \begin{cases} x=x \\ z=\sqrt{1-x^2} \\ y=0 \end{cases} \left\{ \begin{array}{l} dx=dx \\ dz = \frac{-x}{\sqrt{1-x^2}} dx \\ dy=0 \end{array} \right. \left\{ d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k} \right.$$

$$I_1 = \int_{\widehat{DA}} \vec{F} \cdot d\vec{r} = \int_{\widehat{DA}} (xz\vec{i} + x\vec{j} - yz\vec{k}) (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$I_1 = \int_{\widehat{DA}} xzdx + xdy - yzdz$$

$$I_1 = \int_0^1 x \cdot \sqrt{1-x^2} dx - 0 \cdot dz = \int_0^1 x\sqrt{1-x^2} dx$$

$1-x^2 = t$
 $-2x dx = dt$

$$I_1 = -\frac{1}{2} \int_1^0 \sqrt{t} dt = -\frac{1}{2} \cdot \frac{t^{3/2}}{3/2} \Big|_1^0 = -\frac{1}{3} (0-1) = \frac{1}{3}$$

$$* \overline{AB} : \begin{cases} x=x \\ y=1-x \\ z=0 \end{cases} \left\{ \begin{array}{l} dx=dx \\ dy=-dx \\ dz=0 \end{array} \right.$$

$$I_2 = \int_{\overline{AB}} xzdx + xdy - yzdz = \int_1^0 x \cdot (-dx) = -\frac{x^2}{2} \Big|_1^0 = \frac{1}{2}$$

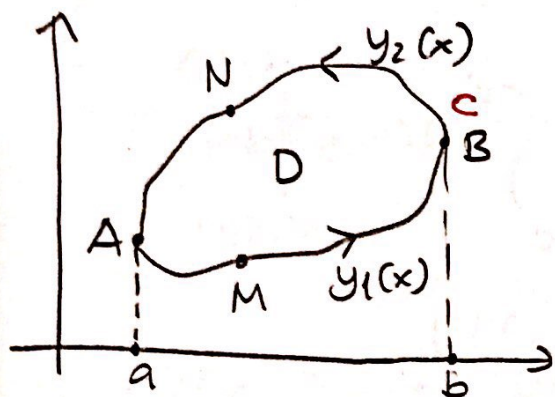
$$* \overline{BC} : \begin{cases} x=0 \\ y=1 \\ z=z \end{cases} \left\{ \begin{array}{l} dx=0 \\ dy=0 \\ dz=dz \end{array} \right.$$

$$I_3 = \int_{\overline{BC}} xzdx + xdy - yzdz = \int_0^1 -1 \cdot z \cdot dz = -\frac{z^2}{2} \Big|_0^1 = -\frac{1}{2}$$

$$I = I_1 + I_2 + I_3 = \frac{1}{3} + \frac{1}{2} - \frac{1}{2} = \frac{1}{3}$$

Green's Formula

Let consider a smooth region D which is bounded by the curves $y=y_1(x)$ and $y=y_2(x)$. Let the functions $P(x,y)$ and $Q(x,y)$ and their partial derivatives be defined and cont. on this region D .



$$I_1 = \iint_D \frac{\partial P(x,y)}{\partial y} dx dy$$

$$I_1 = \int_a^b \int_{y_1(x)}^{y_2(x)} \frac{\partial P(x,y)}{\partial y} dy dx$$

$$I_1 = \int_a^b P(x,y) \Big|_{y_1(x)}^{y_2(x)} dx = \int_a^b [P(x,y_2(x)) - P(x,y_1(x))] dx$$

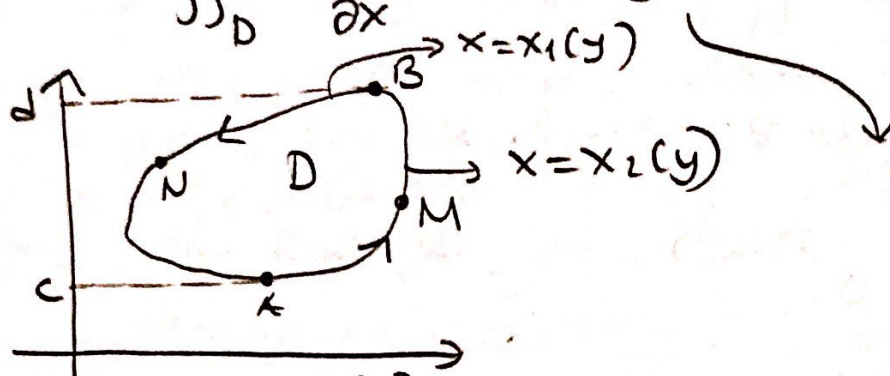
$$I_1 = \underbrace{\int_a^b P(x,y_2(x)) dx}_{\substack{x=x \\ y=y_2(x) \\ \widehat{ANB} \\ \text{line intg.}}} - \underbrace{\int_a^b P(x,y_1(x)) dx}_{\substack{x=x \\ y=y_1(x) \\ \widehat{AMB} \\ \text{line intg.}}}$$

$$I_1 = \int_{\widehat{ANB}} P(x,y) dx - \int_{\widehat{AMB}} P(x,y) dx$$

$$I_1 = - \int_{\widehat{BNA}} P(x,y) dx - \int_{\widehat{AMB}} P(x,y) dx$$

$$I_1 = - \oint_C P(x,y) dx$$

$$I_2 = \iint_D \frac{\partial Q(x, y)}{\partial x} dx dy$$



$$I_2 = \int_c^d \int_{x_1(y)}^{x_2(y)} \frac{\partial Q(x, y)}{\partial x} dx dy = \int_c^d Q(x, y) \Big|_{x_1(y)}^{x_2(y)} dy$$

$$I_2 = \int_c^d (Q(x_2(y), y) - Q(x_1(y), y)) dy$$

$$I_2 = \underbrace{\int_c^d Q(x_2(y), y) dy}_{\text{line integral on } \widehat{A \rightarrow B} \atop x = x_2(y) \atop y = y}} - \underbrace{\int_c^d Q(x_1(y), y) dy}_{\text{line integral on } \widehat{A \rightarrow B}}$$

$$I_2 = \int_{\widehat{A \rightarrow B}} Q(x, y) dy + \int_{\widehat{A \rightarrow B}} Q(x, y) dy$$

$$I_2 = \int_{\widehat{A \rightarrow B}} Q(x, y) dy + \int_{\widehat{B \rightarrow A}} Q(x, y) dy$$

$$I_2 = \oint_C Q(x, y) dy$$

let consider $I_2 - I_1$;

$$I_2 - I_1 = \oint_C P(x, y) dx + Q(x, y) dy$$

$$\oint_C P(x, y) dx + Q(x, y) dy = \iint_D \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx dy$$

Green's Formula

Special Case of Green's Formula;

If $Q(x, y) = x$, $P(x, y) = -y$ then;

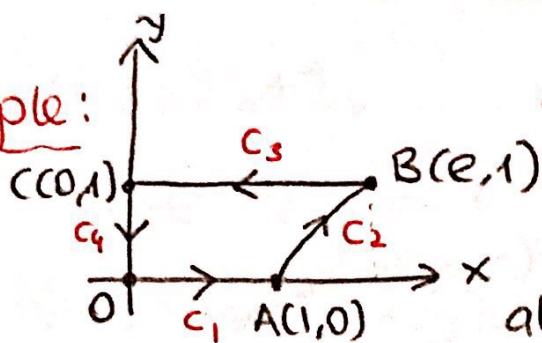
$$\oint_C P(x, y) dx + Q(x, y) dy = \iint_D \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx dy \text{ becomes;}$$

$$\oint_C P(x, y) dx + Q(x, y) dy = \iint_D \underbrace{[1 - (-1)]}_2 dx dy$$

$$\underbrace{\iint_D dx dy}_A = \frac{1}{2} \oint_C -y dx + x dy$$

$$A = \frac{1}{2} \oint_C x dy - y dx = \iint_D dx dy$$

Example:



a) Evaluate

$$I = \oint_C 2dy - ydx$$

also b) apply green formula.

$$C_1: \overline{OA} : \quad x=x \Rightarrow dx=dx \\ y=0 \Rightarrow dy=0$$

$$I_1 = \int_{C_1} 2dy - ydx = \int_0^1 2 \cdot 0 - 0 \cdot dx = 0$$

$$C_2: \widehat{AB} : \quad x=x \Rightarrow dx=dx \\ y=\ln x \Rightarrow dy = \frac{dx}{x}$$

$$I_2 = \int_1^e 2 \cdot \frac{dx}{x} - \ln x dx = \int_1^e \left(\frac{2}{x} - \ln x \right) dx$$

$$I_2 = 2 \ln x - (x \ln x - x) \Big|_1^e = 2 \ln x - x \ln x + x \Big|_1^e$$

$$I_2 = 2 \ln e - e \ln e + e - (2 \cdot 0 - 1 \cdot 0 + 1)$$

$$I_2 = 2 - e + e - 1 = 1$$

$$C_3: \overline{BC} : \quad x=x \Rightarrow dx=dx \\ y=1 \Rightarrow dy=0$$

$$I_3 = \int_{C_3} 2dy - ydx = \int_e^0 2 \cdot 0 - 1 \cdot dx = \int_0^e dx = e$$

$$C_4: \overline{CO} : \quad x=0 \Rightarrow dx=0 \\ y=y \Rightarrow dy=dy$$

$$I_4 = \int_{C_4} 2dy - ydx = \int_1^0 2dy - 0 = 2 \cdot y \Big|_1^0 = -2$$

$$I = I_1 + I_2 + I_3 + I_4$$

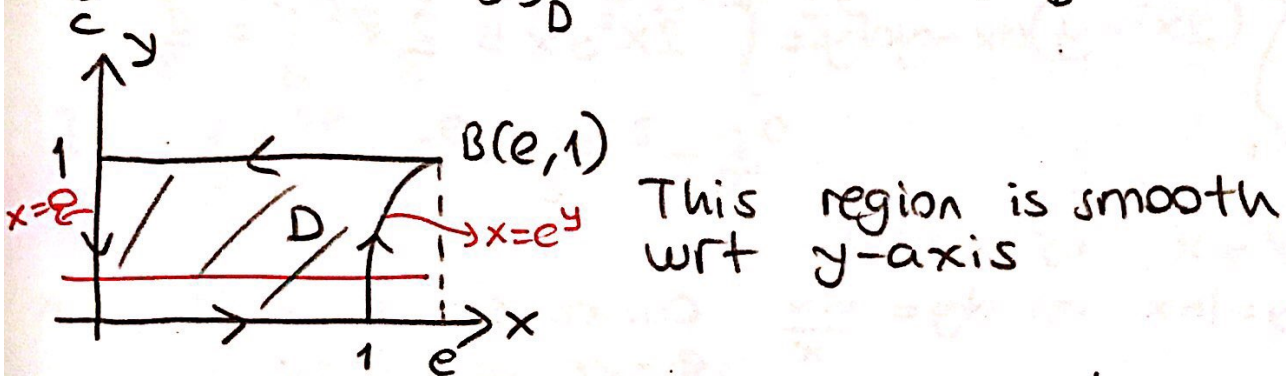
$$I = 0 + 1 + e - 2 = e - 1$$

$$b) \oint_C P(x,y)dx + Q(x,y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$P(x,y) = -y, \quad Q(x,y) = 2$$

$$\frac{\partial P}{\partial y} = -1, \quad \frac{\partial Q}{\partial x} = 0$$

$$\oint_C 2dy - ydx = \iint_D (0 + 1) dx dy = \iint_D dx dy$$



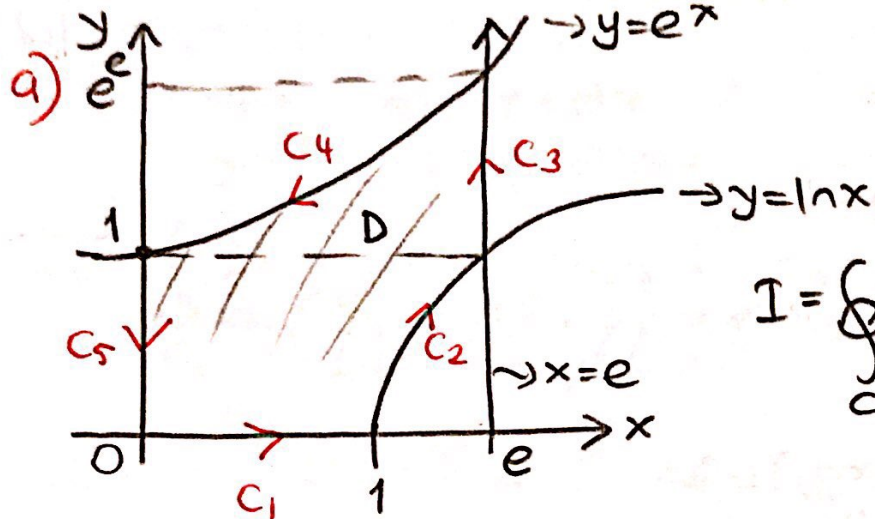
$$I = \int_0^1 \int_0^{e^y} dx dy = \int_0^1 (e^y - 0) dy = e^y \Big|_0^1 = \underline{e - 1}$$

Example: $C: y = \ln x, y = e^x, x = 0, y = 0, x = e$

a) Firstly sketch the region and then evaluate

$$I = \oint_C (2x^2 - y)dx - ydy$$

b) Apply green formula.



$$I = \oint_C (2x^2 - y) dx - y dy$$

* C_1 : $x=x \Rightarrow dx=dx$
 $y=0 \Rightarrow dy=0$

$$I_1 = \int_{C_1} (2x^2 - y) dx - y dy = \int_0^1 2x^2 dx = \frac{2}{3} x^3 \Big|_0^1 = \frac{2}{3}$$

* C_2 : $x=x \Rightarrow dx=dx$
 $y=\ln x \Rightarrow dy = \frac{dx}{x}$

$$I_2 = \int_{C_2} (2x^2 - y) dx - y dy = \int_1^e (2x^2 - \ln x) dx - \ln x \cdot \frac{dx}{x}$$

$$I_2 = \int_1^e (2x^2 - \ln x - \frac{\ln x}{x}) dx = \frac{2}{3} x^3 - (x \ln x - x) - \frac{\ln^2 x}{2} \Big|_1^e$$

$$I_2 = \frac{2}{3} x^3 - x \ln x + x - \frac{\ln^2 x}{2} \Big|_1^e$$

$$I_2 = \frac{2}{3} e^3 - \cancel{e} + \cancel{e} - \frac{1}{2} - \left(\frac{2}{3} + 1 \right) = \frac{2}{3} e^3 - \frac{1}{2} - \frac{2}{3} - 1$$

$$I_2 = \frac{2}{3} e^3 - \frac{13}{6}$$

* C_3 : $x=e \Rightarrow dx=0$
 $y=y \Rightarrow dy=dy$

$$I_3 = \int_{C_3} (2x^2 - y) dx - y dy = \int_e^{e^e} (2e^2 - y) 0 - y dy$$

$$I_3 = \int_1^{e^e} -y dy = -\frac{y^2}{2} \Big|_1^{e^e} = -\frac{e^{2e}}{2} + \frac{1}{2}$$

$$* C_4: \begin{aligned} x=x &\Rightarrow dx=dx \\ y=e^x &\Rightarrow dy=e^x dx \end{aligned}$$

$$I_4 = \int_{C_4} (2x^2 - y)dx - ydy = \int_e^0 (2x^2 - e^x)dx - e^x \cdot e^x dx$$

$$I_4 = \int_e^0 (2x^2 - e^x - e^{2x})dx = \left[\frac{2}{3}x^3 - e^x - \frac{1}{2}e^{2x} \right]_e^0$$

$$I_4 = -e^0 - \frac{1}{2}e^0 - \left(\frac{2}{3}e^3 - e^e - \frac{1}{2}e^{2e} \right)$$

$$I_4 = -1 - \frac{1}{2} - \frac{2}{3}e^3 + e^e + \frac{1}{2}e^{2e}$$

$$I_4 = \frac{1}{2}e^{2e} + e^e - \frac{2}{3}e^3 - \frac{3}{2}$$

$$* C_5: \begin{aligned} x=0 &\Rightarrow dx=0 \\ y=y &\Rightarrow dy=dy \end{aligned}$$

$$I_5 = \int_{C_5} (2x^2 - y)dx - ydy = \int_1^0 -y \cdot 0 - ydy = -\frac{y^2}{2} \Big|_1^0 = \frac{1}{2}$$

$$I = I_1 + I_2 + I_3 + I_4 + I_5$$

$$I = \frac{2}{3} + \cancel{\frac{2}{3}e^3} - \frac{13}{6} - \cancel{\frac{e^{2e}}{2}} + \frac{1}{2} + \cancel{\frac{1}{2}e^{2e}} + \underline{e^e} - \cancel{\frac{2}{3}e^3} - \underline{\frac{3}{2}} + \underline{\frac{1}{2}}$$

$$I = e^e - \frac{1}{2} + \frac{2}{3} - \frac{13}{6}$$

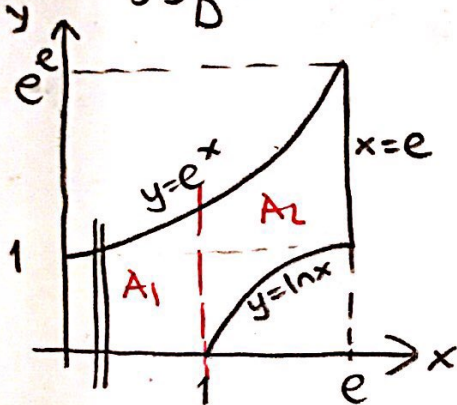
$$\underline{I = e^e - 2}$$

b) Green Theorem;

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_D \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dx dy$$

$$P(x,y) = 2x^2 - y, \quad Q(x,y) = -y$$
$$\frac{\partial P}{\partial y} = -1, \quad \frac{\partial Q}{\partial x} = 0$$

$$I = \iint_D (0 + 1) dx dy = \iint_D dx dy$$



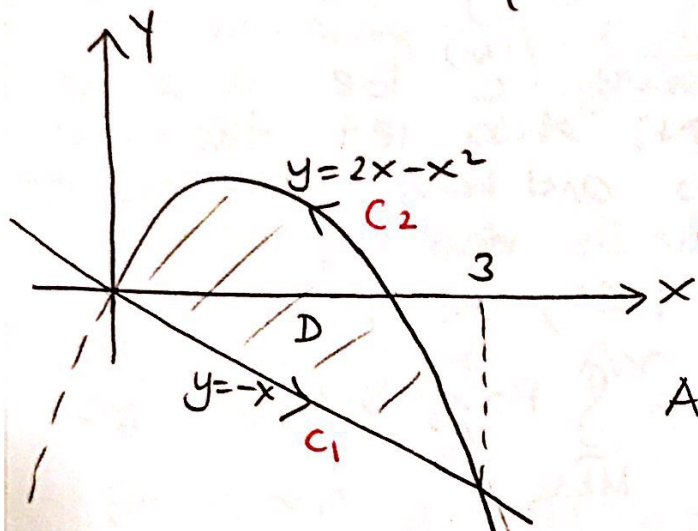
$$I = \int_0^1 \int_0^{e^x} dy dx + \int_1^e \int_{\ln x}^{e^x} dy dx$$

$$I = \int_0^1 e^x dx + \int_1^e (e^x - \ln x) dx = e^x \Big|_0^1 + \left[e^x - (x \ln x - x) \right]_1^e$$

$$I = e - 1 + e^e - \cancel{e \ln e} + e - (e + 1)$$

$$I = \cancel{e - 1} + e^e - \cancel{e - 1} = \underline{e^e - 2}$$

Example: Evaluate the area of the region bounded by the curves $\begin{cases} y = 2x - x^2 \\ y = -x \end{cases}$ by using line integral.



$$-x = 2x - x^2 \Rightarrow x^2 - 3x = 0 \\ x = 0, x = 3$$

$$(*) A = \iint_D dx dy$$

$$A = \int_0^3 \int_{-x}^{2x-x^2} dy dx = \int_0^3 (2x - x^2 + x) dx$$

$$A = \int_0^3 (3x - x^2) dx = \left. \frac{3}{2}x^2 - \frac{x^3}{3} \right|_0^3 = \frac{9}{2}$$

2nd way: $A = \frac{1}{2} \oint_C x dy - y dx$

$$* C_1: \begin{aligned} x &= x \Rightarrow dx = dx \\ y &= -x \Rightarrow dy = -dx \end{aligned}$$

$$I_1 = \int_{C_1} x dy - y dx = \int_0^3 -x dx + x dx = 0$$

$$* C_2: \begin{aligned} x &= x \Rightarrow dx = dx \\ y &= 2x - x^2 \Rightarrow dy = (2 - 2x) dx \end{aligned}$$

$$I_2 = \int_{C_2} x dy - y dx = \int_3^0 x(2 - 2x) dx - (2x - x^2) dx$$

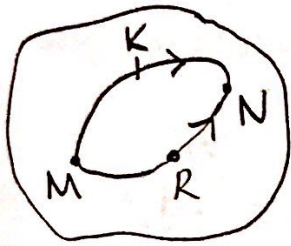
$$I_2 = \int_3^0 (\cancel{2x} - 2x^2 - \cancel{2x} + x^2) dx = \int_0^3 x^2 dx = \left. \frac{x^3}{3} \right|_0^3 = 9$$

$$A = \frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \cdot (0 + 9) = \frac{9}{2}$$

Path Independence for Line Integrals

Consider the line integral $I = \int_{(M)}^{(N)} P(x,y)dx + Q(x,y)dy$

In region D , let the curve $C^{(M)}$ be a path from the point (M) to (N) . Also let the funct. $P(x,y), Q(x,y)$ be continuous and have cont. derivatives. If the line integral is not depending on path (path independent) then;



$$\int_{\widehat{MRN}} Pdx + Qdy = \int_{\widehat{MKN}} Pdx + Qdy$$

$$\int_{\widehat{MRN}} Pdx + Qdy - \int_{\widehat{MKN}} Pdx + Qdy = 0$$

$$\int_{\widehat{MRN}} Pdx + Qdy + \int_{\widehat{NKM}} Pdx + Qdy = 0$$

$$\oint_C Pdx + Qdy = 0$$

Theorem: let the functions $P(x,y), Q(x,y), \frac{\partial Q}{\partial x}, \frac{\partial P}{\partial y}$

be defined and continuous on a region D . Let C be the curve that is enclosing the region D . The line integral

$$\oint_C Pdx + Qdy = 0 \text{ if and only if } \boxed{\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}}$$

holds for all $x, y \in D$.

Proof: Sufficiency $\Leftarrow$: let $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ hold.
By Green's Formula;

$$\oint_C P dx + Q dy = \iint_D \underbrace{\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)}_0 dx dy = 0$$

Necessity $\Rightarrow$: let $I = \oint_C P dx + Q dy = 0$. let $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ do not hold for at least one ^{point} $M(x_0, y_0)$.

$$\frac{\partial P(x_0, y_0)}{\partial y} \neq \frac{\partial Q(x_0, y_0)}{\partial x}$$

Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} > 0$ at M (or $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} < 0$)

$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ is a cont. function so it's positive for all points of a region (small enough) D' including M .

That's for $\epsilon > 0$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} > \epsilon$$

$$\iint_{D'} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy > \iint_{D'} \epsilon dx dy = \epsilon \iint_{D'} dx dy$$

$$= \epsilon \cdot A > 0$$

$\Rightarrow \iint_{D'} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C P dx + Q dy > 0$ but it's a contradiction so $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ must hold.

Result: If $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ holds in a line integral

$I = \int_C P dx + Q dy$ then the result of integration is not depending on the path. (path independent)

If $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ that means the ^{total} differential $P dx + Q dy$ is an exact differential. That's there exists a function $u(x, y)$ such that $P dx + Q dy = du(x, y)$. In this case:

$$I = \int_{(M)}^{(N)} P dx + Q dy = \int_{(M)}^{(N)} du(x, y) = u(x, y) \Big|_{(M)}^{(N)} = \underline{u(N) - u(M)}$$

Note 1: Let C be a space-curve. The line integral

$I = \int_C P(x, y, z) dx + Q(x, y, z) dy + R(x, y, z) dz$ is not depending on the path if the following equations hold;

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}, \quad \frac{\partial R}{\partial x} = \frac{\partial P}{\partial z}.$$

Note 2: $\int \vec{F} d\vec{r} = \int P(x, y) dx + Q(x, y) dy$

$$\vec{F} = P(x, y)\vec{i} + Q(x, y)\vec{j}$$

If $\int \vec{F} \cdot d\vec{r}$ is not depending on the path we have $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ and $P(x, y) dx + Q(x, y) dy = du(x, y)$ is an exact diff.

Then; $\vec{F} = P(x,y)\vec{i} + Q(x,y)\vec{j}$

$$\vec{F} = \frac{\partial u}{\partial x}\vec{i} + \frac{\partial u}{\partial y}\vec{j} = \text{grad } u = \nabla u.$$

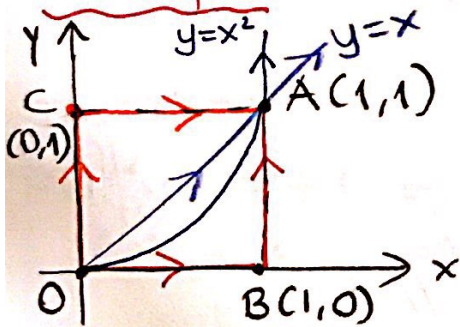
Therefore;

For the line integral $\int \vec{F} d\vec{r}$ to be path-independent, there must exist the function $u = u(x,y)$ such that $\vec{F} = \nabla u$. So we must have $\text{rot } \vec{F} = 0$.

~~Example~~ $\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = 0$

$$\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k} = 0.$$

Example: Evaluate $I = \int_C y^2 x dx + x^2 y dy$;



$$i) I = \int_{\overline{OB}} y^2 x dx + x^2 y dy + \int_{\overline{BA}} y^2 x dx + x^2 y dy$$

$$\overline{OB} : \begin{matrix} y=0 \Rightarrow dy=0 \\ x=x \Rightarrow dx=dx \end{matrix} \left. \vphantom{\begin{matrix} y=0 \\ x=x \end{matrix}} \right\} I_1 = \int_0^1 0 + 0 = 0$$

$$\overline{BA} : \begin{matrix} x=1 \Rightarrow dx=0 \\ y=y \Rightarrow dy=dy \end{matrix} \left. \vphantom{\begin{matrix} x=1 \\ y=y \end{matrix}} \right\} I_2 = \int_0^1 y^2 \cdot 0 + 1 \cdot y dy = \left. \frac{y^2}{2} \right|_0^1 = \frac{1}{2}$$

$$ii) \overline{OA} : \left. \begin{array}{l} y=x \Rightarrow dy=dx \\ x=x \Rightarrow dx=dx \end{array} \right\}$$

$$I = \int_0^1 x^2 \cdot x dx + x^2 \cdot x dx = \int_0^1 2x^3 dx = \frac{2}{4} x^4 \Big|_0^1 = \frac{1}{2} //$$

$$iii) \overline{OA} : \left. \begin{array}{l} y=x^2 \Rightarrow dy=2x dx \\ x=x \Rightarrow dx=dx \end{array} \right\}$$

$$I = \int_0^1 x^4 \cdot x dx + x^2 \cdot x^2 \cdot 2x dx = \int_0^1 \underbrace{(x^5 + 2x^5)}_{3x^5} dx = \frac{3}{6} x^6 \Big|_0^1$$

$$I = \frac{1}{2} //$$

$$iv) \overline{OC} + \overline{CA}$$

$$\overline{OC} : \left. \begin{array}{l} x=0 \Rightarrow dx=0 \\ y=y \Rightarrow dy=dy \end{array} \right\}$$

$$I = \int_0^1 0 + 0 = 0$$

$$\overline{CA} : \left. \begin{array}{l} x=x \Rightarrow dx=dx \\ y=1 \Rightarrow dy=0 \end{array} \right\}$$

$$I = \int_0^1 1 \cdot x dx + x^2 \cdot 1 \cdot 0 = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2} //$$



$$(*) I = \int_C y^2 x dx + x^2 y dy, \quad \begin{array}{l} P(x,y) = y^2 x \\ Q(x,y) = x^2 y \end{array}$$

$$\left(\frac{\partial P}{\partial y} = 2xy \right) = \left(\frac{\partial Q}{\partial x} = 2xy \right) \Rightarrow \text{integral is not depending on the path!}$$

So there exists a function $u=u(x,y)$ such that $P(x,y) = \frac{\partial u}{\partial x}$ and $Q(x,y) = \frac{\partial u}{\partial y}$

$$\frac{\partial u}{\partial x} = y^2 x \Rightarrow u(x, y) = \frac{x^2 y^2}{2} + h(y)$$

$$\frac{\partial u}{\partial y} = x^2 y + \frac{dh}{dy} = x^2 y \Rightarrow h = c \text{ (const.)}$$

$$u = u(x, y) = \frac{x^2 y^2}{2} + c$$

$$\int P(x, y) dx + Q(x, y) dy = \int du(x, y) = u(x, y)$$

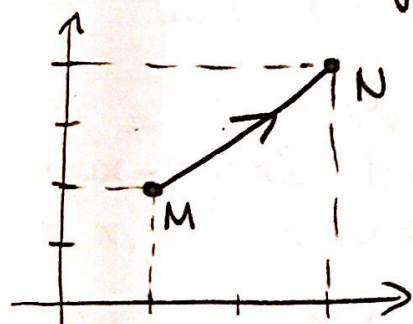
$$\int y^2 x dx + x^2 y dy = \left. \frac{x^2 y^2}{2} + c \right|_{(0,0)}^{(1,1)} = \left(\frac{1}{2} + c \right) - (0 + c) = \frac{1}{2} //$$

Example: Evaluate

$$I = \int_{(M)} (6xy^2 - y^3) dx + (6x^2y - 3xy^2) dy$$

M(1, 2), N(3, 4).

Is this integral depending on the path?



M(1, 2), N(3, 4)

$$\frac{x-1}{1-3} = \frac{y-2}{2-4} \Rightarrow y = x + 1$$

$$\overline{MN}: \left. \begin{array}{l} y = x + 1 \Rightarrow dy = dx \\ x = x \Rightarrow dx = dx \end{array} \right\}$$

$$I = \int_1^3 [6x(x+1)^2 - (x+1)^3] dx + [6x^2(x+1) - 3x(x+1)^2] dx$$

$$I = \int_1^3 [6x(x^2 + 2x + 1) - (x^3 + 3x^2 + 3x + 1) + 6x^3 + 6x^2 - 3x(x^2 + 2x + 1)] dx$$

$$I = \int_1^3 (6x^3 + 12x^2 + 6x - x^3 - 3x^2 - 3x - 1 + 6x^3 + 6x^2 - 3x^3 - 6x^2 - 3x) dx$$

$$I = \int_1^3 (8x^3 + 9x^2 - 1) dx = \left. \frac{8}{4}x^4 + \frac{9}{3}x^3 - x \right|_1^3 = 236$$

$$(*) \quad I = \int_{(M)}^{(N)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy$$

$$P(x,y) = 6xy^2 - y^3, \quad Q(x,y) = 6x^2y - 3xy^2$$

$$\frac{\partial P}{\partial y} = 12xy - 3y^2$$

$$\frac{\partial Q}{\partial x} = 12xy - 3y^2$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \text{integral is not depending on the path.}$$

So we can find a function $u = u(x,y)$ s.t.

$$\frac{\partial u}{\partial x} = P(x,y) = 6xy^2 - y^3, \quad \frac{\partial u}{\partial y} = Q(x,y) = 6x^2y - 3xy^2$$

$$u(x,y) = 3x^2y^2 - y^3x + h(y)$$

$$\frac{\partial u}{\partial y} = 6x^2y - 3y^2x + \frac{dh}{dy} = 6x^2y - 3xy^2 \Rightarrow h = C \text{ constant}$$

$$u(x,y) = 3x^2y^2 - xy^3 + C$$

$$I = \int_{(M)}^{(N)} ((6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy) = \int_{(M)}^{(N)} d(3x^2y^2 - xy^3 + C)$$

$$I = 3x^2y^2 - xy^3 + C \Big|_{(1,2)}^{(3,4)} = 236 //$$

Example: Evaluate $I = \int_{(A)}^{(B)} (\arctan y - 2x)dx +$

$(\frac{x}{1+y^2} + 2y)dy$ on the way of the line

between the points $A(1,1)$, $B(2,\sqrt{3})$.

$$\frac{x-1}{1-2} = \frac{y-1}{1-\sqrt{3}} \Rightarrow (1-\sqrt{3})(x-1) = 1-y$$

$$y = (\sqrt{3}-1)x + 1-\sqrt{3}+1$$

$$y = (\sqrt{3}-1)x + 2-\sqrt{3}$$

$$\overline{AB} : \quad x=x \quad \Rightarrow dx=dx$$

$$y=(\sqrt{3}-1)x+2-\sqrt{3} \quad \Rightarrow dy=(\sqrt{3}-1)dx$$

$$I = \int_1^2 \arctan((\sqrt{3}-1)x+2-\sqrt{3}) dx + \left(\frac{x}{1+(-\dots)^2} + 2(-\dots)^2 \right) (\sqrt{3}-1) dx$$

Is this integral depending on the path?

$$P(x,y) = \arctan y - 2x, \quad Q(x,y) = \frac{x}{1+y^2} + 2y$$

$$\frac{\partial P}{\partial y} = \frac{1}{1+y^2}, \quad \frac{\partial Q}{\partial x} = \frac{1}{1+y^2}$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \text{integral is path independent.}$$

So there exists a function $u(x,y)$ s.t.;

$$\frac{\partial u}{\partial x} = \arctan y - 2x \quad \text{and} \quad \frac{\partial u}{\partial y} = \frac{x}{1+y^2} + 2y$$

$$u(x,y) = x \cdot \arctan y - x^2 + h(y)$$

$$\frac{\partial u}{\partial y} = \frac{x}{1+y^2} + \frac{dh}{dy} = \frac{x}{1+y^2} + 2y \Rightarrow h(y) = y^2 + c$$

$$u(x,y) = x \cdot \arctan y - x^2 + y^2 + c$$

$$I = \int_{(A)}^{(B)} (\arctan y - 2x) dx + \left(\frac{x}{1+y^2} + 2y \right) dy = \int_{(A)}^{(B)} d(x \arctan y - x^2 + y^2 + c)$$

$$I = x \cdot \arctan y - x^2 + y^2 + c \Big|_{(1,1)}^{(2,\sqrt{3})}$$

$$I = 2 \cdot \overbrace{\arctan \sqrt{3}}^{\pi/3} - 4 + 3 + c - (1 \cdot \overbrace{\arctan 1}^{\pi/4} - 1 + 1 + c)$$

$$I = 2 \frac{\pi}{3} - 1 - \frac{\pi}{4} \Rightarrow I = \frac{5\pi}{12} - 1 //$$

Example: Evaluate $I = \int (e^{x+z})dx + \ln(1+z)dy + \left(\frac{y}{1+z} + x\right)dz$

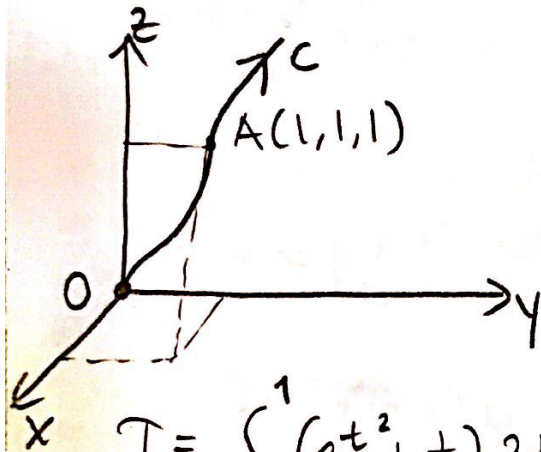
on the curve $\vec{r}(t) = t^2\vec{i} + t^3\vec{j} + t\vec{k}$ between the points $O(0,0,0)$ and $A(1,1,1)$.

Parametric equations of the curve;

$$x = t^2 \Rightarrow dx = 2t dt$$

$$y = t^3 \Rightarrow dy = 3t^2 dt$$

$$z = t \Rightarrow dz = dt$$



$$\text{for } O(0,0,0) \rightarrow t=0$$

$$\text{for } A(1,1,1) \rightarrow t=1$$

$$I = \int_0^1 (e^{t^2+t}) 2t dt + \ln(1+t) 3t^2 dt + \left(\frac{t^3}{1+t} + t^2 \right) dt$$

$$I = \int_0^1 e^{t^2} \cdot 2t dt + \int_0^1 2t^2 dt + 3 \int_0^1 t^2 \ln(1+t) dt + \int_0^1 \frac{t^3}{1+t} dt + \int_0^1 t^2 dt$$

$$I = \int_0^1 e^{t^2} \cdot 2t dt + \frac{2}{3} t^3 \Big|_0^1 + 3 \int_0^1 t^2 \ln(1+t) dt + \int_0^1 \frac{t^3}{1+t} dt + \frac{t^3}{3} \Big|_0^1$$

$\begin{matrix} u=t^2 \\ du=2t dt \end{matrix} \quad \begin{matrix} \ln(1+t)=u \\ \frac{dt}{1+t}=du \end{matrix} \quad \begin{matrix} t^2 dt=dv \\ v=\frac{t^3}{3} \end{matrix}$

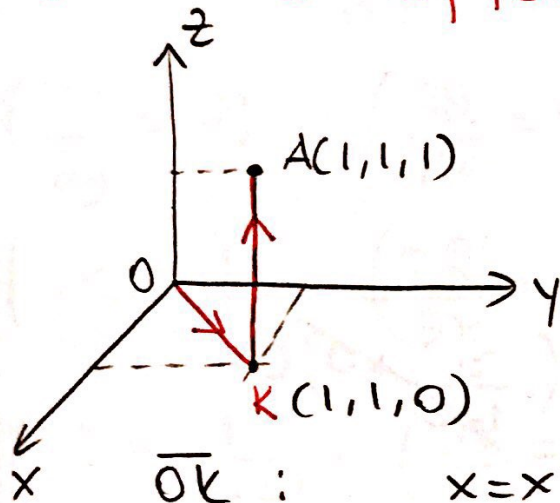
$$I = \int_0^1 e^u du + \frac{2}{3} + \frac{1}{3} + \cancel{\frac{t^3}{3} \cdot \ln(1+t) \Big|_0^1} - \cancel{3 \cdot \int_0^1 \frac{t^3}{3} \cdot \frac{dt}{1+t}} + \int_0^1 \left(t^2 - t + 1 - \frac{1}{1+t} \right) dt$$

$$I = e^u \Big|_0^1 + 1 + 1 \cdot \ln 2 - \int_0^1 \frac{t^3}{1+t} dt + \int_0^1 \frac{t^5}{1+t} dt \quad [19]$$

$$I = e - 1 + 1 + \ln 2$$

$$\underline{I = e + \ln 2}$$

Choose a different path;



$$\overline{OK} + \overline{KA} : c$$

$$\overline{OK} : \begin{cases} x=x \\ y=x \\ z=0 \end{cases} \begin{cases} dx=dx \\ dy=dx \\ dz=0 \end{cases}$$

$$I_1 = \int_0^1 (e^x + 0) dx + \underbrace{\ln(1+0)}_0 \cdot dx + 0 = e^x \Big|_0^1 = e - 1$$

$$\overline{KA} : \vec{u} = \langle 0, 0, 1 \rangle \quad \begin{cases} x = 1 + 0 \cdot t \\ y = 1 + 0 \cdot t \\ z = 0 + t \end{cases} \begin{cases} x=1 \rightarrow dx=0 \\ y=1 \rightarrow dy=0 \\ z=t \rightarrow dz=dt \end{cases}$$

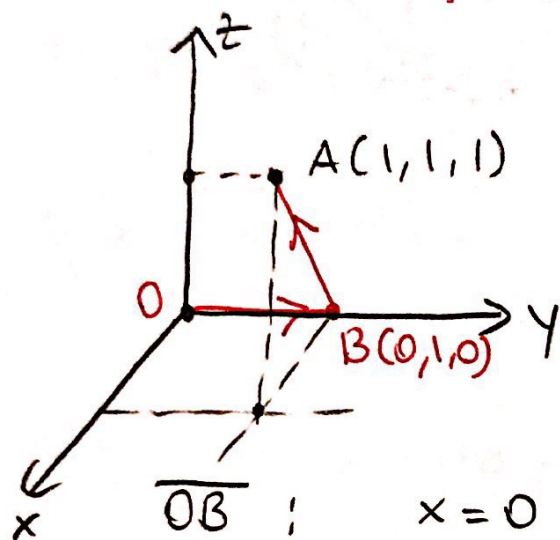
$$I_2 = \int_0^1 (e+t) 0 + \ln(1+t) \cdot 0 + \left(\frac{1}{1+t}\right) dt = \int_0^1 \left(\frac{1}{1+t} + 1\right) dt$$

$$I_2 = \ln(1+t) + t \Big|_0^1 = \ln 2 + 1 - 0 = \ln 2 + 1$$

$$I = I_1 + I_2 = e - 1 + \ln 2 + 1 = \underline{e + \ln 2}$$

Choose a different path;

$$\overline{OB} + \overline{BA}$$



$$\overline{OB} : \begin{cases} x=0 \\ y=y \\ z=0 \end{cases} \Rightarrow \begin{cases} dx=0 \\ dy=dy \\ dz=0 \end{cases}$$

$$I_1 = \int_0^1 (e^0 + 0)0 + \ln(1+0)dy + \left(\frac{y}{1+0} + 0 \right)0$$

$$I_1 = 0$$

$$\overline{BA} : \vec{u} = \langle -1, 0, -1 \rangle, B(0,1,0)$$

$$\begin{aligned} x = -t &\Rightarrow dx = -dt & \begin{cases} \text{for } (0,1,0) \rightarrow t=0 \\ \text{for } (1,1,1) \rightarrow t=-1 \end{cases} \\ y = 1 &\Rightarrow dy = 0 \\ z = -t &\Rightarrow dz = -dt \end{aligned}$$

$$I_2 = \int_0^{-1} (e^{-t} - t)(-dt) + \ln(1-t) \cdot 0 + \left(\frac{1}{1-t} - t \right)(-dt)$$

$$I_2 = \int_0^{-1} \left(t - e^{-t} + t - \frac{1}{1-t} \right) dt$$

$$I_2 = \left[t^2 + e^{-t} + \ln(1-t) \right]_0^{-1}$$

$$I_2 = (1 + e + \ln 2) - (0 + 1 + 0)$$

$$\underline{I_2 = e + \ln 2} \Rightarrow \underline{I = I_1 + I_2 = e + \ln 2}$$

Is the integral path-independent?

$$I = \int_C (e^x + z)dx + \ln(1+z)dy + \left(\frac{y}{1+z} + x\right)dz$$

$$P(x, y, z) = e^x + z, \quad Q(x, y, z) = \ln(1+z)$$

$$R(x, y, z) = \frac{y}{1+z} + x$$

$$\left(\frac{\partial P}{\partial y} = 0\right) = \left(\frac{\partial Q}{\partial x} = 0\right) \checkmark$$

$$\left(\frac{\partial P}{\partial z} = 1\right) = \left(\frac{\partial R}{\partial x} = 1\right) \checkmark$$

$$\left(\frac{\partial Q}{\partial z} = \frac{1}{1+z}\right) = \left(\frac{\partial R}{\partial y} = \frac{1}{1+z}\right) \checkmark$$

integral is not depending on the path.

So, there exists a function $u(x, y, z)$ such that

$$\frac{\partial u}{\partial x} = e^x + z, \quad \frac{\partial u}{\partial y} = \ln(1+z), \quad \frac{\partial u}{\partial z} = \frac{y}{1+z} + x$$

$$u(x, y, z) = e^x + z \cdot x + h(y, z)$$

$$\frac{\partial u}{\partial y} = \frac{\partial h}{\partial y} = \ln(1+z) \Rightarrow h(y, z) = y \cdot \ln(1+z) + \alpha(z)$$

$$u(x, y, z) = e^x + zx + y \ln(1+z) + \alpha(z)$$

$$\frac{\partial u}{\partial z} = x + \frac{y}{1+z} + \frac{d\alpha}{dz} = \frac{y}{1+z} + x \Rightarrow \alpha = C \text{ (constant)}$$

$$u(x, y, z) = e^x + zx + y \ln(1+z) + C$$

$$I = \int_{(0,0,0)}^{(1,1,1)} d(e^x + zx + y \ln(1+z) + C)$$

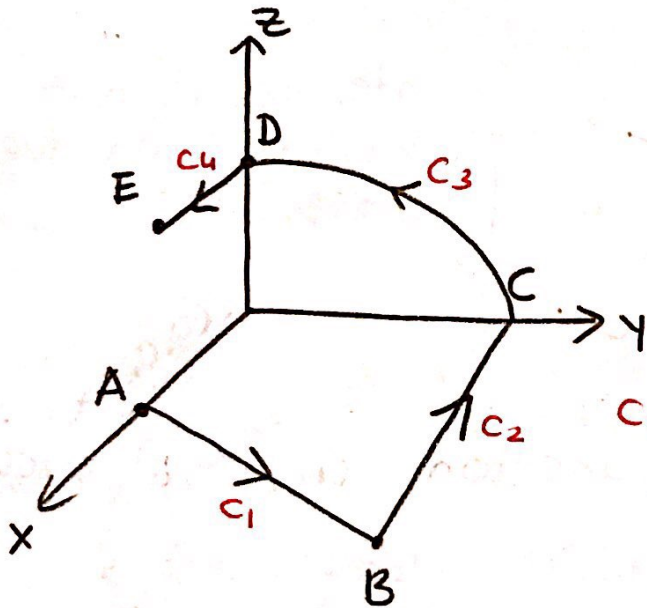
$$I = e^x + xz + y \ln(1+z) + C \quad \Big|_{(1,1,1)}^{(0,0,0)}$$

$$I = e + 1 + 1 \cdot \ln 2 + C - (1 + 0 + 0 + C)$$

$$I = e + \cancel{1} + \ln 2 + \cancel{C} - \cancel{1} - \cancel{C}$$

$$\underline{I = e + \ln 2}$$

Example: Evaluate $I = \int_C ((3x^2 + 3y - 1)dx + (z^2 + 3x)dy + (1 + 2yz)dz)$



$$A(2,0,0), B(1,3,0)$$

$$C(0,4,0), D(0,0,4)$$

$$E(-1,-1,2)$$

$$C_1: \overline{AB} \text{ line}$$

$$\frac{x-2}{2-1} = \frac{y-0}{0-3}$$

$$y = -3x + 6$$

$$\left. \begin{array}{l} x=x \rightarrow dx=dx \\ y=-3x+6 \rightarrow dy=-3dx \\ z=0 \rightarrow dz=0 \end{array} \right\}$$

$$I_1 = \int_2^1 (3x^2 + 18 - 9x - 1)dx + (0 + 3x)(-3dx) + 0$$

$$I_1 = \int_2^1 (3x^2 - 18x + 17)dx = x^3 - 9x^2 + 17x \Big|_2^1$$

$$I_1 = 1 - 9 + 17 - (8 - 36 + 34) = 9 - 6$$

$$\underline{I_1 = +3}$$

(2)

C_2 : line $\overline{BC}$: $\frac{x-1}{1-0} = \frac{y-3}{3-4} \Rightarrow x-1 = 3-y$
 $y = 4-x$

$$\left. \begin{array}{l} x=x \\ y=4-x \\ z=0 \end{array} \right\} \begin{array}{l} dx=dx \\ dy=-dx \\ dz=0 \end{array}$$

$$I_2 = \int_0^1 (3x^2 + 12 - 3x - 1)dx + (0 + 3x)(-dx) + 0$$

$$I_2 = \int_1^0 (3x^2 - 3x + 11 - 3x)dx = \int_1^0 (3x^2 - 6x + 11)dx$$

$$I_2 = x^3 - 3x^2 + 11x \Big|_1^0 = 0 - (1 - 3 + 11) = -9 //$$

C_3 : $\widehat{CD}$: $y^2 + z^2 = 16$
 $x=0$

$$\left. \begin{array}{l} y^2 + z^2 = 16 \\ x=0 \end{array} \right\} \begin{array}{l} dx=0 \\ y=4\cos\theta \\ z=4\sin\theta \end{array} \left\{ \begin{array}{l} dy=-4\sin\theta d\theta \\ dz=4\cos\theta d\theta \end{array} \right.$$

$$I_3 = \int_0^{\pi/2} (0 + 12\cos\theta - 1) \cdot 0 + (16\sin^2\theta + 0) \cdot (-4\sin\theta d\theta) + (1 + 2 \cdot 16 \cdot \sin\theta \cos\theta) \cdot 4\cos\theta d\theta$$

$$I_3 = \int_0^{\pi/2} (-64\sin^3\theta + 4\cos\theta + 8 \cdot 16 \cdot \sin\theta \cos^2\theta) d\theta$$

$$I_3 = -64 \int_0^{\pi/2} \sin\theta \cdot (1 - \cos^2\theta) d\theta + 4 \sin\theta \Big|_0^{\pi/2} + 128 \int_0^{\pi/2} \sin\theta \cos^2\theta d\theta$$

$\cos\theta = u$
 $-\sin\theta d\theta = du$

$$I_3 = +64 \int_0^1 (1 - u^2) du + 4 - 128 \int_0^1 u^2 du$$

$$I_3 = 64 \cdot \left[u - \frac{1}{3}u^3 \right] \Big|_0^1 + 4 - \frac{128}{3} \cdot u^3 \Big|_0^1$$

$$I_3 = 64 \left(-1 + \frac{1}{3} \right) + 4 + \frac{128}{3} = 4 + \frac{128}{3} + \frac{64 \cdot (-2)}{3}$$

$I_3 = 4$

C_4 : line $\overline{DE}$: $D(0,0,4)$, $E(-1,-1,2)$

$$\vec{v} = \langle -1, -1, -2 \rangle$$

$$\left. \begin{array}{l} x = -t \\ y = -t \\ z = 4 - 2t \end{array} \right\} \begin{array}{l} dx = -dt \\ dy = -dt \\ dz = -2dt \end{array} \left\{ \begin{array}{l} \text{for } D(0,0,4) \rightarrow t=0 \\ \text{for } E(-1,-1,2) \rightarrow t=1 \end{array} \right.$$

$$I_4 = \int_0^1 (3t^2 - 3t - 1)(-dt) + ((4-2t)^2 - 3t)(-dt) + (1 + 2(-t)(4-2t))(-2dt)$$

$$I_4 = \int_0^1 (-3t^2 + 3t + 1 + 3t + 16t - 16 - 4t^2) dt + (-2 + 4t(4-2t)) dt$$

$$I_4 = \int_0^1 (-7t^2 + 22t - 15 - 2 + 16t - 8t^2) dt$$

$$I_4 = \int_0^1 (-15t^2 + 38t - 17) dt$$

$$I_4 = (-5t^3 + 19t^2 - 17t) \Big|_0^1 = -5 + 19 - 17$$

$$\underline{I_4 = -3}$$

$$I = I_1 + I_2 + I_3 + I_4$$

$$I = \cancel{2} + (-9) + 4 - \cancel{3} \Rightarrow \boxed{I = -5}$$

Is this integral path-independent?

$$I = \int_C \underbrace{(3x^2 + 3y - 1)}_P dx + \underbrace{(z^2 + 3x)}_Q dy + \underbrace{(1 + 2yz)}_R dz$$

$$\frac{\partial P}{\partial y} = 3$$

$$\frac{\partial Q}{\partial x} = 3 \quad \checkmark$$

$$\frac{\partial Q}{\partial z} = 2z$$

$$\frac{\partial R}{\partial y} = 2z \quad \checkmark$$

$$\frac{\partial P}{\partial z} = 0$$

$$\frac{\partial R}{\partial x} = 0 \quad \checkmark$$

$\Rightarrow$ path-independent.

Then, there exists function $u(x, y, z)$ such that

$$\frac{\partial u}{\partial x} = 3x^2 + 3y - 1, \quad \frac{\partial u}{\partial y} = z^2 + 3x, \quad \frac{\partial u}{\partial z} = 1 + 2yz$$

$$u(x, y, z) = x^3 + 3xy - x + h(y, z)$$

$$\frac{\partial u}{\partial y} = 3x + \frac{\partial h}{\partial y} = z^2 + 3x \Rightarrow \frac{\partial h}{\partial y} = z^2$$

$$h(y, z) = z^2 y + \alpha(z)$$

$$u(x, y, z) = x^3 + 3xy - x + z^2 y + \alpha(z)$$

$$\frac{\partial u}{\partial z} = 2zy + \frac{d\alpha}{dz} = 1 + 2yz \Rightarrow \alpha = z + C$$

$$u(x, y, z) = x^3 + 3xy - x + z^2 y + z + C$$

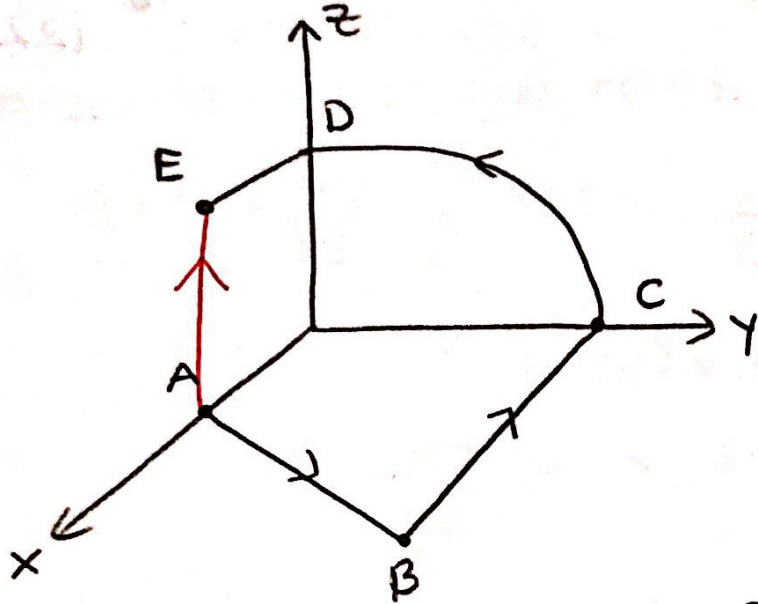
$$I = u \Big|_{(A)}^{(E)} = x^3 + 3xy - x + z^2 y + z + C \Big|_{(2, 0, 0)}^{(-1, -1, 2)}$$

$$I = -1 + 3 + 1 - 4 + 2 + C - (8 - 2 + C)$$

$$I = 1 - 6$$

$$I = -5$$

⊗ Since the integral is not depending on the path, then we can choose a shorter way between A and E.



line $\overline{AE}$;

$$\vec{v} = \langle -3, -1, 2 \rangle$$

$$A(2, 0, 0)$$

$$x = 2 - 3t \Rightarrow dx = -3dt$$

$$y = -t \Rightarrow dy = -dt$$

$$z = 2t \Rightarrow dz = 2dt$$

$$\text{for } A(2, 0, 0) \rightarrow t = 0$$

$$\text{for } E(-1, -1, 2) \rightarrow t = 1$$

$$I = \int_0^1 (3(2-3t)^2 - 3t - 1)(-3dt) + (4t^2 + 3(2-3t))(-dt) + (1 + 2(2t)(-t))2dt$$

$$I = \int_0^1 (3(4-12t+9t^2) - 3t - 1)(-3dt) + (-4t^2 - 6 + 9t)dt + (2 - 8t^2)2dt$$

$$I = \int_0^1 (12 - 36t + 27t^2 - 3t - 1)dt + (-4t^2 - 6 + 9t + 2 - 8t^2)dt$$

$$I = \int_0^1 (-81t^2 + 117t - 33 - 12t^2 + 9t - 4)dt$$

$$I = \int_0^1 (-93t^2 + 126t - 37)dt$$

$$I = -31t^3 + 63t^2 - 37t \Big|_0^1$$

$$I = -31 + 63 - 37$$

$$\boxed{I = -5}$$